

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2023-24

Paper: BSC-04A — Elements of Geology - II

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 074447

*Instructions: Answer **five** questions in all, selecting at least **two** questions from each of Sections A and B. Section-C is compulsory. All questions carry equal marks.*

SECTION — A

Question 1. Draw a geological time scale marking the boundaries of EON and ERA in million years. Explain the evolution of life through different geological periods. [14]

Question 2. Answer the following:

- (a) How is the physiography of the Northern Mountains different from that of the Peninsular Plateaus? [7]
- (b) What are Steno's Laws and how do they help in establishing the relative dating of different stratigraphic units? [7]

Question 3. Answer the following:

- (a) Discuss the differences between lithostratigraphic and biostratigraphic units. [5]
- (b) Discuss the desertification of Thar and mention the characteristic topography. [5]
- (c) Comment on the 'Great Dying Event'. [4]

Question 4. Answer the following:

- (a) Name the eastern and western coastal plains of India. [4]
- (b) Explain the Law of Uniformitarianism. [4]
- (c) Describe the division of the Northern Plains. [3]
- (d) Explain the physiographic divisions of India. [3]

SECTION — B

Question 5. Discuss the Siwalik vertebrate fauna and describe their significance in paleoclimatic interpretations. [14]

Question 6. Define a fossil. Describe different modes of occurrence of fossils. [14]

Question 7. Write explanatory notes on the following:

- (a) Conditions for fossilization. [6]
- (b) Scope of Palaeontology. [4]
- (c) Index fossil and its use. [4]

SECTION — C

Question 8. Define the following in a maximum of five sentences each: [2 × 7 = 14]

- (i) Unconformity
- (ii) Adam's Bridge
- (iii) Minicoy Island
- (iv) Kankar
- (v) Living fossil
- (vi) Fossil in toto
- (vii) Silicification